

Original Report

Quantification of the impact of multifaceted initiatives intended to improve operational efficiency and the safety culture: A case study from an academic medical center radiation oncology department

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Abstract

Purpose: We have systematically been incorporating several operational efficiency and safety initiatives into our academic radiation oncology clinic. We herein quantify the impact of these initiatives on prospectively collected, clinically meaningful, metrics.

Methods and Materials: The data from 5 quality improvement initiatives, each focused on a specific safety/process concern in our clinic, are presented. Data was collected prospectively: operational metrics recorded before and after implementation of the initiative were compared using statistical analysis. Results from the Agency for Health Care Research and Quality (AHRQ) patient safety culture surveys administered during and after many of these initiatives were similarly compared.

Results: (1) Workload levels for nurses assisting with brachytherapy were high (National Aeronautics and Space Administration Task Load Index (NASA-TLX) scores >55-60, suggesting, “overwork”). Changes in work flow and procedure room layout reduced workload to more acceptable levels (NASA-TLX <55; $P < .01$). (2) The rate of treatment therapists being interrupted was reduced from a mean of 4 (range, 1-11) times per patient treatment to a mean <1 (range, 0-3; $P < .001$) after implementing standards for electronic communication and placement of monitors informing patients and staff of the treatment machine status (ie, delayed, on time). (3) The rates of replans by dosimetrists was reduced from 11% to 6% ($P < .01$) through a more systematic pretreatment peer review process. (4) Standardizing nursing and resident functions reduced patient wait times by $\approx 45\%$ (14 min; $P < .01$). (5) Standardizing presimulation instructions from the physician reduced the number of patients experiencing delays on the

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simulator (>50% to <10%; $P < .01$). To assess the overall changes in “patient safety culture,” we conducted a pre- and postanalysis using the AHRQ survey. Improvements in all measured dimensions were noted.

Conclusions: Quality improvement initiatives can be successfully implemented in an academic radiation oncology department to yield measurable improvements in operations resulting in improvement in patient safety culture.

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Introduction

Recent articles in *The New York Times* have highlighted the importance of patient safety in radiation oncology.^{1–4} This publicity has led to several safety-focused meetings sponsored by the Food and Drug Administration, the American Society for Radiation Oncology, and the American Association of Physicists in Medicine, emphasizing the importance of improving the quality of patient care in radiation oncology. Recognizing the importance of improving patient safety in radiation oncology, since 2009 we have implemented several quality improvement initiatives that are amenable to metrics that quantify their implementation and associated outcome measures. We herein provide quantitative and qualitative data reflecting the incorporation of some of these initiatives in our clinic and how these initiatives have improved the patient safety culture in our department.

Methods and materials

From 2009 to 2011 a series of initiatives were planned and implemented within the department of radiation oncology. Various improvement methodologies and tools were used to spearhead improvement efforts. For several of these initiatives, prospective quantitative data were collected in order to assess if the initiative was operating as intended (ie, measuring the *process*), and/or assess if the initiative was having the desired result (ie, measuring the *outcome*). A brief description of each assessed initiative and associated measure(s) is described below. To assess concurrent changes in “patient safety culture,” we evaluated Agency for Healthcare Research and Quality (AHRQ) Medical Office Surveys on Patient Safety Culture.

Quality initiative #1: Attempts to reduce the workload of nurses performing high-dose-rate radiation implant procedures

Suboptimal workload (over- and underload) has been correlated with increased errors in several industries, including medicine.^{5,6} We previously measured the workload for different radiation oncology professionals in our department using the National Aeronautics and Space Administration task load index (NASA-TLX).⁷ NASA-TLX is a subjective, multidimensional assessment

tool that quantifies perceived workload and has been used in multiple settings such as aviation. We spent 46 hours (15 brachytherapy procedures) formally observing the nursing process, performed a hierarchical task analysis to divide the procedures into various tasks, and collected NASA-TLX scores on the identified tasks. A systematic human error reduction and prediction approach (SHERPA) was used to classify potential errors related to high-workload tasks⁸ and changes to processes were implemented to reduce workload and stressors.

Quality initiative #2: Changes in processes aimed to reduce interruptions of radiation therapists during patient treatments

In 2010 we observed that radiation therapists working on our linear accelerators were interrupted an average of 4 times per patient treatment. The preponderance of interruptions were patient phone calls to the machine inquiring whether they were “on time” and nonurgent communication between other radiation oncology professionals and the radiation therapist. We implemented practical solutions aimed to reduce interruptions and reassessed the interruption rate. Interventions included rerouting of phones, placing nonurgent communication between radiation oncology professionals in the electronic medical record system (Mosaik; Elekta, Stockholm, Sweden), and the institution of public display of the treatment machine status (ie, on time vs delayed).

Quality initiative #3: Enhancement of pretreatment clinical peer review as a component of a daily morning huddle

Peer review has been shown to be feasible in radiation oncology centers and provides quality assurance of radiation therapy (RT) treatment planning resulting in improvement in patient care.⁹ We increased the formality of our long-standing pre-RT peer review processes in order to both improve the quality of radiation treatment plans and reduce the number replans. All patients are discussed prior to initiation of therapy, typically after computed tomographic (CT) simulation but prior to treatment planning. We review the indications for treatment, image segmentation (particularly for IMRT [intensity modulated RT] cases), proposed dose and fractionation, and

discuss tentative beam arrangements (for non-IMRT 3D [3-dimensional] cases). Attendees typically include faculty physicians, dosimetrists, residents, nurses, physicists, therapists, students, and administrators. Further, this peer review was integrated as the central component of a daily departmental “morning huddle” where the day’s schedule is discussed (ie, brachytherapy and simulation procedures), that day’s anticipated challenges, inpatient census, and where all sections of the department (ie, dosimetry, therapy, physics, nursing) are able to raise concerns, make announcements, etc.

We also conduct a second peer review of patients during their first week of treatment that is akin to traditional “chart rounds.” This meeting is usually attended by a subset of the same staff that attends pre-RT peer review. Attention is paid to the dosimetry, portal films, and dose prescriptions.

In order to assess the utility of the pre-RT peer review and morning huddle, we measured the percent of cases with suggested changes (somewhat arbitrarily segregated into minor and major [see Results]), the number of patients who initiated therapy then needed a replan (excluding planned field reductions or changes indicated for medical reasons such as tumor progression), and surveyed departmental personnel who attended the meeting. Furthermore, we also monitored the number of changes instituted based on the second “on-treatment” peer review session as an indication of the utility of the pre-RT review session.

Quality initiative #4: Changing processes to compensate for inefficiencies introduced with the transition from a paper-chart to an electronic medical record system

In approximately 2008 to 2009, our clinic completed its transition to an electronic medical record system (Mosaik) that includes scheduling and clinic support, replacing the previously used paper chart. However, the paper chart served functions that were not readily provided by Mosaik. For example, the chart served as a visual cue to prompt staff that patients were ready for movement between “stations” in the clinic (ie, waiting room, simulation, treatment machines, and clinic exam room). Loss of this visual cue led to disrupted flow, long wait times, and nursing rework and overwork. Over a 3-day period, researchers and staff with specialized training in operations performed detailed observations of clinical operations. We identified multiple opportunities for altering processes to reduce wasted motion, wait times, and better utilize staff. We instituted a multipronged system including the following: (a) a central white board and color-coded plastic sleeves identifying patient location by physician and arrival time; (b) a flag system at each exam room, alerting providers to waiting patients; and (c) communication boards at each exam room, alerting providers to clinical details of waiting patients. We hired a unit coordinator to disseminate limited, key

information from Mosaik to the white board and assist with communications. We quantified the impact of these initiatives on clinic staff satisfaction (survey), clinical flow (measured patient’s wait time prior to being put into a clinic room), and nursing work (measured the time it took to perform a nursing assessment; ie, time from patient being placed in clinic room to when the first provider was alerted), through measurements before and after the initiatives.

Quality initiative #5: Reducing delays in CT simulation

Starting in approximately 2009, we performed a series of formal operational analyses and observations of clinical operations focusing on patient flow in the simulator. Causes of delays were identified and procedures were modified accordingly. For example, methods of communicating information to and from the simulator (ie, physician instructions to the therapists, appointment times, and instructions to the patients) were standardized and formalized. Further, review of the daily simulator schedule was performed each morning as part of the morning huddle (see above); cases where physician instructions are lacking or ambiguous are highlighted, and anticipated challenges are discussed.

Evaluation of patient safety culture

The AHRQ defines the safety culture of an organization “... as the product of individual and group values, attitudes, perceptions, competencies, and patterns of behavior that determine the commitment to, and the style and proficiency of, an organization’s health and safety management.”¹⁰ Creating a patient safety culture is a critical component of any type of safety improvement program. Outside of health care, a strong safety culture is noted as a critical underpinning in other high reliability organizations.¹⁰⁻¹³ AHRQ has an assessment tool that measures the patient safety culture for hospitals, nursing homes, ambulatory outpatient medical offices, and pharmacies. The AHRQ patient safety survey measures 12 dimensions of patient safety culture. Our institution routinely administers this survey every 18 months for all outpatient clinics. We compared survey results from 2009 to 2011 to examine trends in patient safety culture change over time and to evaluate the cultural impact associated with the aforementioned patient safety initiatives and interventions.

Statistics

Descriptive statistics are used to present the data. Where possible, comparisons were made using the Student *t* test and analysis of variance.

Table 1 Human factors improvement to reduce workload for brachytherapy nurses

Human factors improvement	Result
1. Medication table was moved next to the patient's bed	→ Improved efficiency/workflow
2. Additional instruments were purchased to improve efficiency.	→ Prior to this only 1 set was available and had to be cleaned prior to each HDR procedure, causing significant delays and interruptions.
3. Moved glove box from the sink to work table/patients bed	→ Improved efficiency/workflow
4. Sterile gowns and caps were moved to provide more direct access	→ Improved efficiency/workflow
5. An additional, portable examination light was placed in the treatment room	→ Improved efficiency. Previously 1 light was shared between multiple providers limiting productivity.
6. Phone numbers at the nurses desk in the HDR suite and at the patient's bedside were changed to carry the same number	→ Reduced rework. Prior to this change the nurse had to tend to 2 phones.
7. All HDR cables are now tethered together	→ Improved efficiency and safety. Prior to this change, often the nurse would have to search for a missing catheter. Also tethering them together has reduced inadvertent disconnection from the HDR unit and brachytherapy apparatus.

HDR, high dose rate.

Results

Quality initiative #1: Quantitative assessment of workload in radiation oncology

The following tasks performed by the HDR nurses scored the highest level of workload ($P < .0001$): 1) CT scan and radiation delivery preparation (59.9; SD, 13.6); (2) Patient identification and preparation (60.6; SD, 9.25); (3) cervix preparation (58.1; SD, 10.2); and (4) physician assistance and outreach (59.8; SD, 15.1). The SHERPA analysis found the following human errors: (1) Information miscommunication (ie, incorrect vital signs); and (2) inappropriately conducted or missed tasks (ie, incorrect placement of Foley catheter and rectal tube, etc). These errors did not result in patient harm but did cause rework and consequentially unnecessary stress and frustration.

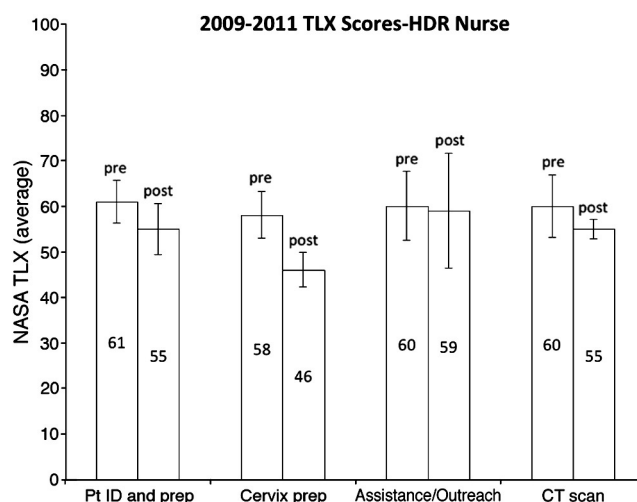


Figure 1 Impact of improvement initiatives on the workload level for the nurses performing high-dose-rate (HDR) brachytherapy procedures. National Aeronautics and Space Administration task load index (NASA-TLX) scores for HDR Nurses before (2009) and after (2011) interventions were implemented to standardize or streamline workflow, remove nonvalue added tasks (see Table 1), and improve the HDR room layout. The tasks listed on the x axis are those identified as having the highest workload levels. The workload levels postintervention are reduced compared with the preintervention levels and are at acceptable levels (TLX <55). (ID, identification; CT, computed tomography.)

Major changes were made to standardized procedures and to improve the physical layout of the HDR treatment room (Table 1). The leadership team worked closely with the HDR nurses over a period of 16 months to change the workflow and treatment room layout. Table 1 lists examples of human factors engineering improvements. Figure 1 shows the NASA-TLX scores before and after the interventions; $P < .01$.

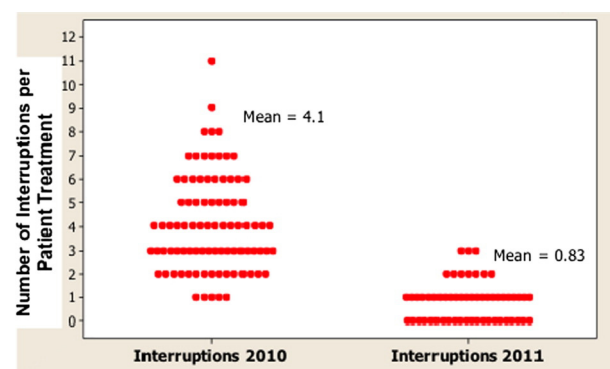


Figure 2 Impact of operational changes on the number of interruptions of radiation therapists on treatment machine. Each dot represents an observation of a single patient's treatment. In 2010, therapists were interrupted on average 4 times per patient treatment. Several policies were implemented to reduce interruptions, and on repeat measurement the average number of interruptions was decreased to <1 ($P < .001$).

Table 2 Rate of suggested changes in the radiation treatment plans during peer review

Results of peer review	Pretreatment planning review (977 cases)	Posttreatment planning, on-treatment review ("chart rounds") (1005 cases)
No change	23%	94%
Minor change ^a	26%	5%
Major change ^b	51%	1%

For the pretreatment peer review major changes were defined as treating the wrong area, ≥ 1 -cm field edge or target contour alteration, and change in the number of sites to be treated. A minor change^a was defined as < 1 -cm field edge or target contour alteration. Changes in intended prescription or treatment technique were not classified as a major or minor change. For posttreatment peer review, major change^b was defined as change that resulted in a replan (ie, exceeding normal tissue tolerances, inadequate target coverage, incorrect dose/fractionation, etc). Minor changes were defined as altering patient setup based on review of portal imaging. It should be noted that not all suggested changes were implemented.

Quality initiative #2: Interruptions of radiation therapists during patient treatments

The pre- and postintervention data are shown in Fig 2. The mean number of interruptions was reduced from 4 (range, 0-11) to < 1 (range, 0-3); $P < .001$.

Quality initiative #3: Clinical peer review

The majority of patients have changes suggested during pre-RT peer review (Table 2).

Smaller fractions (8%) of patients have changes suggested at the second "on treatment" review, perhaps reflecting the effectiveness of the pretreatment peer review process. The fraction of patients who initiate therapy then need a replan (excluding planned field reductions or changes indicated for medical reasons) over time is shown in Fig 3; $P < .01$. A survey of departmental personnel who attend the pretreatment peer review and morning huddle shows broadly favorable responses (Table 3).

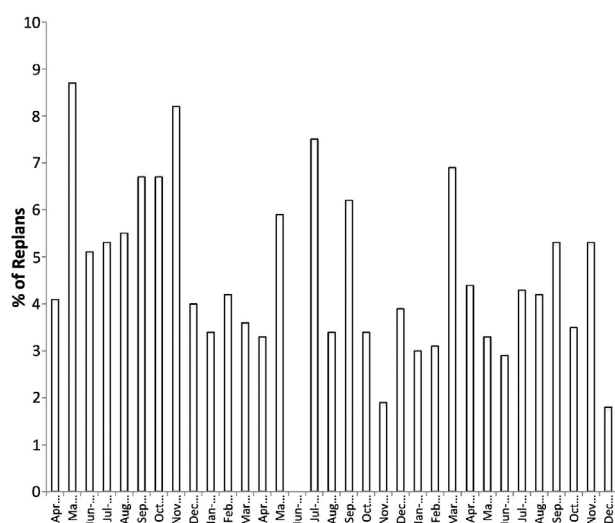


Figure 3 Rate of replans at University of North Carolina-Chapel Hill over time. Replans were defined as replanning approved plans; ie, patients having a second plan generated by dosimetry, after an initial plan was already approved by the physician.

Quality initiative #4: Changing processes to adjust to the electronic medical record system

After the initial switch to MOSAIQ, only 16% of responding clinicians and nurses (3 of 19) were satisfied with the overall patient flow in the clinic. After implementation of several initiatives (ie, white board to track patients in the clinic) satisfaction increased to 85% (17 of 19). Patient wait times (time from registering at the front desk to being placed in a clinic room) were reduced from a mean of 31 to 17 minutes; $P < .01$. Nursing time for a nurse to prepare new patient consults for the provider was reduced from 28 to 20 minutes (ie, the time between the patients leaving the waiting room to when they were ready to see the provider); $P < .01$.

A initiative #5: Reducing delays in CT simulation

Upon initial assessment of 84 patient delays in the CT simulator, identified causes were generally attributed to physicians ($n = 35$), patients ($n = 32$), nurses ($n = 21$), and other issues ($n = 20$). Table 4 lists major reasons for the delays. The root causes for the majority of delays were related to suboptimal communication. Subsequently, we instituted a mandatory presimulation communication through MOSAIQ. Attending and resident physicians enter in simulation orders via the "QuickRx" function prior to simulation for the therapists. We also instituted other forms of electronic communication via MOSAIQ for the schedulers, front desk personnel, nurses, and simulator therapists to improve patient transfers. The number of patient delays on the CT simulator decreased from $> 50\%$ to $< 10\%$; $P < .01$.

Evaluation of patient safety culture

Figure 4 shows the AHRQ patient safety culture survey results in our department in 2009 and 2011; 60% of our staff participated. Percentages of positive responses for all 12 dimensions of patient safety culture appear to have increased from 2009 to 2011. The surveys are anonymous and staffing changes occurred during this time period; 1 therapist and 2 nurses left our department. Therefore, we

Table 3 Survey results of regular attendees of pretreatment peer review

Question	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
1. Convenient time of day for me			2 7.4%	16 59.3%	9 33.3%
2. Collegial debate/conversation			1 3.7%	19 70.4%	7 25.9%
3. Provides me with quality/value			4 14.8%	15 55.6%	8 29.6%
4. Adds departmental quality/value			1 3.7%	14 51.9%	12 44.4%
5. An excellent learning environment	1 3.7%		4 14.8%	12 44.4%	10 37%
6. Improves departmental "safety climate"			1 3.7%	16 59.3%	10 37%
7. Helps clinical operations run smoothly			3 11.1%	16 59.3%	8 29.6%
8. Fosters communication within the department			1 3.7%	16 59.3%	10 37%
9. I am satisfied with simulation review				20 74.1%	7 25.9%

Surveys were administered to 27 regular attendees of pretreatment peer review including attending and resident physicians, physicists, dosimetrist, administrators, therapy/dosimetry students, and medical students.

do not know to what degree the same groups of people were represented in the survey.

Discussion

Since 2008 we have implemented and quantified the impact of several operational efficiency and safety initiatives in our clinic. Based on our subjective assessments and the quantitative data shown, these initiatives appear to have been, at least in part, successful. Concurrent with these efforts we recognize an apparent improvement in the patient safety culture in our department as measured by an independent AHRQ patient safety culture survey.

The present analysis has several shortcomings. First, the methods used to gather data are imperfect. Surveys, limited clinical observations, and subjective assessments of workload may not be ideal means to objectively assess the impact of the initiatives. Second, in many instances there were other changes to the clinical operation (ie, different staffing over time) that may have influenced the changing metrics. Third, there is a clear reporting bias. We specifically selected to describe several initiatives that have quantitative data suggesting a positive outcome. Many other initiatives had outcomes that were not readily quantifiable and others had clearly suboptimal outcomes. Even when an initiative has an apparent positive outcome, the benefits are not always sustained. For example, despite early improvements in our clinical flow (initiative #4), we experienced intermittent increased challenges in our clinical flow related to staff turnover and inconsistent adherence to defined policies.

Many of the quality initiatives directly improved operational efficiencies and were not directly implemented to address a specific safety issue. However, improving efficiency should reduce workload and resultantly improve patient safety. Workload has been linked to performance and safety in high reliability industries such as aviation and nuclear power plants.¹⁰ In medicine, excessive workload

Table 4 Causes of delays at the computed tomography simulator

No. of delays = 84	Reasons
35 Attributed to physicians	→ No simulation orders → Consent not obtained → Order for contrast administration, pregnancy test was not communicated
32 Attributed to patients	→ Late to appointment → Questions about insurance → Claustrophobia
21 Attributed to nurses	→ Incomplete consent forms → Not informing simulator therapist that patient is ready → Starting intravenous for contrast administration
20 Miscellaneous	→ Waiting on Spanish interpreter → Transportation from inpatient unit → Front desk not informing simulator therapist that patient is ready → Schedulers not telling patient to come 1 hour prior to simulation if patient is receiving contrast

Attribution listed is based on the perceived major determinant of the delay. This is somewhat arbitrary as causes are often multifactorial.

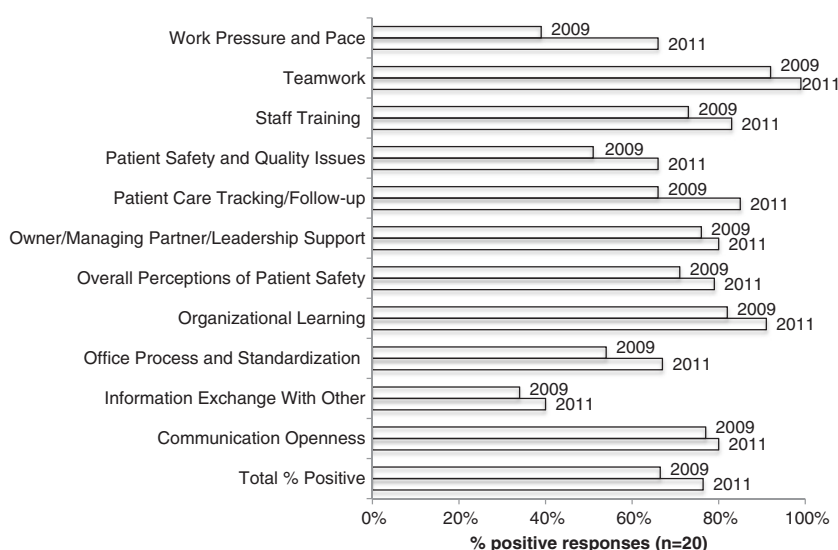


Figure 4 Results from the Agency for Health Care Research and Quality (AHRQ) patient safety culture survey results: 2009 and 2011.

(eg, interruptions) affects performance and compromises patient safety.^{6,7,14} In this paper we have highlighted some of our efforts to reduce workload; limiting interruptions of the therapists and reducing replans. Replans are both inefficient and are often done in a rushed setting. Indeed, one of the events highlighted in *The New York Times* articles involved a replan that was rushed and failed to go through the regular quality assurance procedures.^{4,15} When people are faced with apparently inefficient systems, their tendency is to create workarounds, and to cut corners. These acts tend to reduce standardization and increase risks. If systems are designed as efficiently as possible, then there is less incentive for such workarounds, and corner cutting, and systems are more likely to remain stable.

One of the biggest obstacles we faced was physician buy-in. Physician leadership is a key element to designing, fostering, and nurturing a culture of safety.¹⁶⁻¹⁸ Physician engagement in quality improvement is fundamental for successful organizational cultural change.¹⁹ Physicians must be identified as champions of safety and be involved (from the beginning) in quality initiatives.¹⁹ It is often difficult to get physicians interested in quality and safety as they perceive they are already practicing safely and see quality initiatives as adding to their workload. To get successful buy-in from physicians, quality initiatives should reduce hassles and wasted time and be easy to try and do by the physicians.^{19,20}

Our analysis suggests that a multifaceted approach to address safety, efficiency, and reliability might improve the safety culture in a clinical radiation oncology department. Such effort is often frustrating, time consuming, and requires consistent oversight. Change is hard,²⁰⁻²² staff are often interested in maintaining existing systems that are familiar and that do not require them to alter

their perceptions or actions. Nevertheless, the quantitative data reported suggest that such efforts can be fruitful. Physician and staff leadership are key elements to designing, fostering, and nurturing improvements in the safety culture.¹⁶

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